Technische Universität Chemnitz

Exploring Reinforcement learning strategies in automated wheel truing

Master's Thesis

submitted by

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Abstract

This thesis attempts to find a solution to automatic spoked wheel truing through reinforcement learning in a simulated environment. We test and evaluate a set of different algorithms, state-spaces and reward schemes.

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Chapter 1

Introduction

1.1 Motivation

short motivation preferably relating the problem to material stress optimization problems.

1.2 Wheel truing

describing the problem and mention sim-to-real outlook

1.3 Thesis Structure

This thesis is structured as follows: Chapter 2 reviews related work.

Chapter 2

Theory

This chapter should outline the necessary theoretic background on deep-reinforcment-learning.

2.1 Reinforcement learning basics

Here I want to cover the basic ideas, definitions and mathematical prerequisites. This will probably include:

- Markov states
- states, actions and their values
- value iteration and policies
- function approximation and deep-learning
- data sampling and replay buffer

After these basics, I want to describe the main ideas and various improvements of the different approaches of the three selected algorithms in the next subsections:

2.1.1 DeepQ (Rainbow)

2.1.2 Policy gradients (PPO)

2.1.3 World-Models (TDMPC)

Chapter 3

Environment

Chapter 4

Experiments and Methodology

Chapter 5

Results and Analysis

Chapter 6

Discussion and Conclusion

Appendix

.1 Additional Figures

Additional figures that didn't fit in the main chapters.

.2 Additional Tables

Additional tables with detailed data.

.3 Code Listings

Extended code listings that are too long for the main text.

.4 Experimental Details

Detailed experimental configurations and parameters.